OPEN-SOURCE SOFTWARE CAPSTONE PROJECT

Open-SourceAudit of Linux administration

Submitted by

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Course: Open-Source Software

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Part A: Origin and Philosophy

A1. The problem this software was created to solve

Developed by Guido van Rossum in the late 1980s and officially deployed in 1991, Python was introduced during a period dominated by lower-level languages such as C and C++. While computationally performant, these environments presented steeplearning curves and significant systemic complexity.

The primary limitation of the era was the high syntactic overhead required to execute rudimentary operations. Software engineers were frequently compelled to prioritize strict syntactical rules over algorithmic problem-solving, thereby hindering rapid prototyping and overall accessibility. Python was architected to mitigate these bottlenecks by implementing:

- Highly readable, semantically clear syntax
- An expedited developer onboarding trajectory
- High-level architectural abstractions

The overarching design philosophy prioritized developer ergonomics. By enforcing structural simplicity, Python significantly reduced cognitive load, facilitating more efficient code authoring and long-term maintenance.

Furthermore, Python was built upon an open-source distribution model. This strategic decision fostered a

decentralized ecosystem, enabling the global developer community to iteratively enhance
the language's core repository and standard library over time. Currently, Python serves as foundational architecture across diverse computing domains, including:

- Server-side web frameworks
- Artificial Intelligence and machine learning
- Quantitative data science
- System and task automation

Consequently, Python successfully abstracted systemic complexity while retaining robust computational flexibility and extensibility.

A2. The license-what it actually says

Python is distributed under the Python Software Foundation (PSF) License, which is a permissive open-source license.

Four Freedoms of Open Source

The four essential freedoms are:

1. Freedom to run the program
2. Freedom to study how it works
3. Freedom to modify it
4. Freedom to distribute copies

Python provides all four freedoms, making it a fully pensource software

Can Companies Modify and Sell It?
Yes, companies can modify Python and use it commercially. They are not required to share their changes publicly. This makes the PSF license flexible and business-friendly.

A3. The ethics of open source
Open-source software is not just a technical model but a philosophy that promotes transparency, collaboration, and equal access to knowledge. It allows anyone to use, study, modify, and share software freely, which encourages innovation and learning across the world. By removing barriers such as cost and restricted access, open source creates opportunities for students, developers, and organizations to grow without limitations. It helps build a global community where people contribute together to improve technology.

However, open source also raises important ethical questions. One major concern is whether it is fair for companies to build profitable products using software developed by a community of unpaid contributors. While open source enables businesses to innovate faster and reduce development costs, there is an ethical responsibility for such companies to give back to the community. This contribution can be in the form of improving the code, supporting developers, or funding

open-source initiatives. At the same time, not all software can be fully open due to security, privacy, or business constraints, which makes the issue more complex.

The idea of “standing on the shoulders of giants” is central to open source, as developers build upon existing work to create new and improved solutions. This approach does not reduce originality but instead accelerates progress and innovation. Overall, open source represents a balance between freedom and responsibility, where collaboration must be supported by fairness, contribution, and ethical decision-making.

Part B: Linux Footprint

Python is widely used on Linux systems and is often pre-installed in many distributions. It is commonly used for scripting, automation, and software development, making it an important tool in the Linux environment.

Installation Method

Python can be installed using package managers like apt in Debian-based systems. The installation is simple and ensures all dependencies are handled automatically.
sudo apt update sudo apt install python3

Key Directories

After installation,Python is stored in different system directories:

- /usr/bin/python3 — main executable
- /usr/lib/python3.x/ — libraries
- /usr/local/lib/ — user-installed packages

These directories help organize Python components efficiently.

User and Execution

Python runs under the currently logged-in user and does not require a separate system user. This improves security and gives users control over execution.

Service Management

Python is not a system service. It runs manually using commands like:

python3 file.py

However, it can be automated using tools like cron jobs.

Update Model

Python is updated through the Linux package manager and community contributions.

Updates include bug fixes, security patches, and performance improvements.

Part C: Ecosystem Mapping (Python)

a vast and interconnected open-source ecosystem. Its functionality, growth, and widespread adoption are supported by numerous other open-source tools, libraries, and communities. This interconnected nature highlights one of the key strengths of open-source software collaboration and shared development.

Dependencies of Python:

Python relies on several underlying system components and libraries to function effectively. These dependencies are essential for its performance, security, and compatibility across platforms. Some key dependencies include:

- **GCC (GNU Compiler Collection):** Used for compiling Python from source
- **OpenSSL:** Provides secure communication features such as encryption
- **zlib:** Used for data compression
- **libc (C Standard Library):** Core system library required by most programs

These dependencies demonstrate how Python builds upon other opensource technologies. It reflects the layered nature of software development, where each tool depends on previously developed systems.

What Python Has Enabled:

Python has significantly influenced modern software development by enabling the creation of numerous frameworks, tools, and technologies. Its simplicity and flexibility have made it a preferred choice for developers across various domains.

Some major areas where Python has had a strong impact include:

- **Web Development:** Frameworks like Django and Flask allow developers to build scalable web applications.
- **Data Science and Analytics:** Libraries such as NumPy, Pandas, and Matplotlib are widely used for data analysis.
- **Artificial Intelligence and Machine Learning:** Tools like TensorFlow and PyTorch rely heavily on Python.
- **Automation and Scripting:** Python is commonly used for automating repetitive tasks in Linux systems.

These examples show that Python is not just a programming language but a foundation for many modern technologies.

Community and Governance:

Python is supported by a strong global community and is governed by the Python Software Foundation. This organization is responsible for managing Python's development, maintaining its licensing, and supporting community initiatives.

The Python community contributes through:

- Open-source repositories (GitHub)
- Mailing lists and discussion forums
- Conferences such as PyCon
- Documentation and tutorials

One of the key strengths of Python's ecosystem is its transparent and community-driven governance model. Decisions are made collaboratively, and contributions are encouraged from developers around the world.

Contribution Process

When people want to contribute to Python they do it through a website called GitHub. They send in what is called a request, which is like a suggestion for how to make Python better. Then some other developers look at it. Decide if it is a good idea or not. If it is they add it to the Python project.

Bug Reporting

If someone finds a bug, in Python they report it using something called an issue tracker. Then all the developers work together to fix the bug quickly as they can. This way Python stays working and people can keep using it to make new things.

Part D: Comparative Analysis

Python vs Proprietary Software

Open source and proprietary software represent two different approaches to software development and distribution.

Python, being an open-source programming language, can be compared with proprietary alternatives such as Microsoft's development tools or other closed-source platforms.

This comparison helps in understanding the strengths and limitations of both models.

Advantages of Python

1. Easy to learn and use
2. There are a lot of people who use Python. They help each other
3. Python is free and open source
4. Python can be used for things

Disadvantages of Python

1. Python is slower than some languages
2. Python uses memory than some other languages
3. Python is not the choice for making things for mobile devices

Security and Transparency

Python offers high transparency because its source code is publicly available. Developers can review the code,

identify vulnerabilities, and contribute improvements. This leads to faster detection and fixing of security issues.
On the other hand, proprietary software does not provide access to source code. Users must rely on the company to ensure security, which may limit trust and flexibility.

Scripts

Script 1: System Identity Report

```
==========================================
vardan@DESKTOP-PPK06AU:~/oss_project$ nano script1.sh
vardan@DESKTOP-PPK06AU:~/oss_project$ ./script1.sh
==========================================
Open Source Audit -- Vardan
==========================================
OS         : Ubuntu 24.04.4 LTS
Kernel     : 6.6.87.2-microsoft-standard-WSL2
User       : vardan
Uptime     : up 35 minutes
Date/Time  : Tuesday, 31 March 2026 15:48:27
==========================================
```

Script 2: Package Inspector

```
==========================================
vardan@DESKTOP-PPK06AU:~/oss_project$ nano script2.sh
vardan@DESKTOP-PPK06AU:~/oss_project$ chmod +x script2.sh
vardan@DESKTOP-PPK06AU:~/oss_project$ ./script2.sh
Package [bash] is installed.
----------------------------------
Status: install ok installed
Version: 5.2.21-2ubuntu4
Description: GNU Bourne Again SHell
----------------------------------
Bash: the shell that powers Linux interaction -- GPL v3
```

Script 3: Dictionary and Permission Auditor

```
vardan@DESKTOP-PPK06AU:~/oss_project$ nano script3.sh
vardan@DESKTOP-PPK06AU:~/oss_project$ chmod +x script3.sh
vardan@DESKTOP-PPK06AU:~/oss_project$ ./script3.sh
========================================
   Directory Audit Report
========================================
Directory : /etc
Permissions : drwxr-xr-x
Owner : root | Group : root
Size : 4.3M
----------------------------------
Directory : /home
Permissions : drwxr-xr-x
Owner : root | Group : root
Size : 56K
----------------------------------
Directory : /tmp
Permissions : drwxrwxrwt
Owner : root | Group : root
Size : 24K
----------------------------------
Directory : /usr/bin
Permissions : drwxr-xr-x
Owner : root | Group : root
Size : 95M
----------------------------------

Kernel Module Directory Check:
Found : /lib/modules/6.6.87.2-microsoft-standard-WSL2
drwxr-xr-x 1 root root 40 Mar 31 15:12 /lib/modules/6.6.87.2-microsoft-standard-WSL2
```

Script 4: Log File Analyzer

```
vardan@DESKTOP-PPK06AU:~/oss_project$ nano sample.log
vardan@DESKTOP-PPK06AU:~/oss_project$ nano script4.sh
vardan@DESKTOP-PPK06AU:~/oss_project$ chmod +x script4.sh
vardan@DESKTOP-PPK06AU:~/oss_project$ ./script4.sh sample.log error
Analyzing : sample.log
Keyword   : error
----------------------------------
Keyword 'error' found 3 time(s)

Last matching lines:
Error: failed to load module
Error: disk not found
Error: network timeout
```

Script 5: The Open Source Manifesto Generator

```
vardan@DESKTOP-PPK06AU:~/oss_project$ nano script5.sh
vardan@DESKTOP-PPK06AU:~/oss_project$ chmod +x script5.sh
vardan@DESKTOP-PPK06AU:~/oss_project$ ./script5.sh
========================================
   Open Source Manifesto Generator
========================================
Enter tool you use: Linux
What does freedom mean to you? innovation
What will you build? drone software

Manifesto saved in manifesto.txt
THE OPEN SOURCE MANIFESTO
Generated: 31 March 2026 | Author: vardan
---------------------------------
I use Linux daily.
Freedom means innovation.
I will build drone software and share it openly.
vardan@DESKTOP-PPK06AU:~/oss_project$
```